

OccuPYcy : un package Python pour ajuster des **modèles d'occupancy** sur des **données massives**

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Supervision: Alexis Joly, Christophe Botella, Maximilien Servajean



Inria



Pl@ntNet

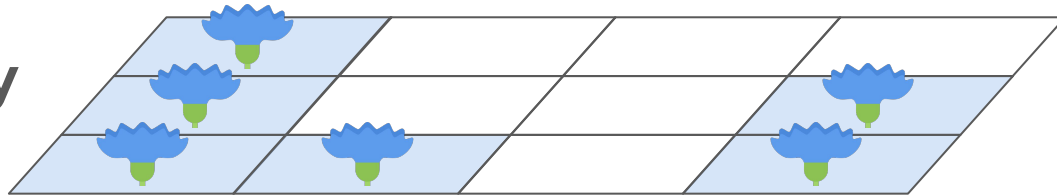


LIRMM

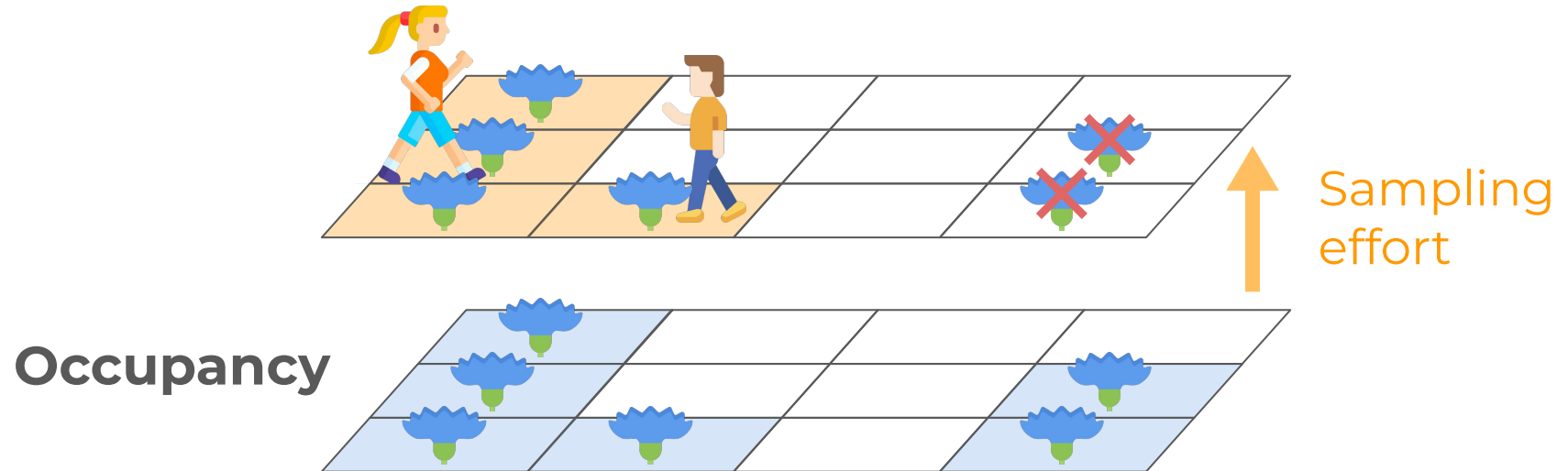
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Modeling occupancy

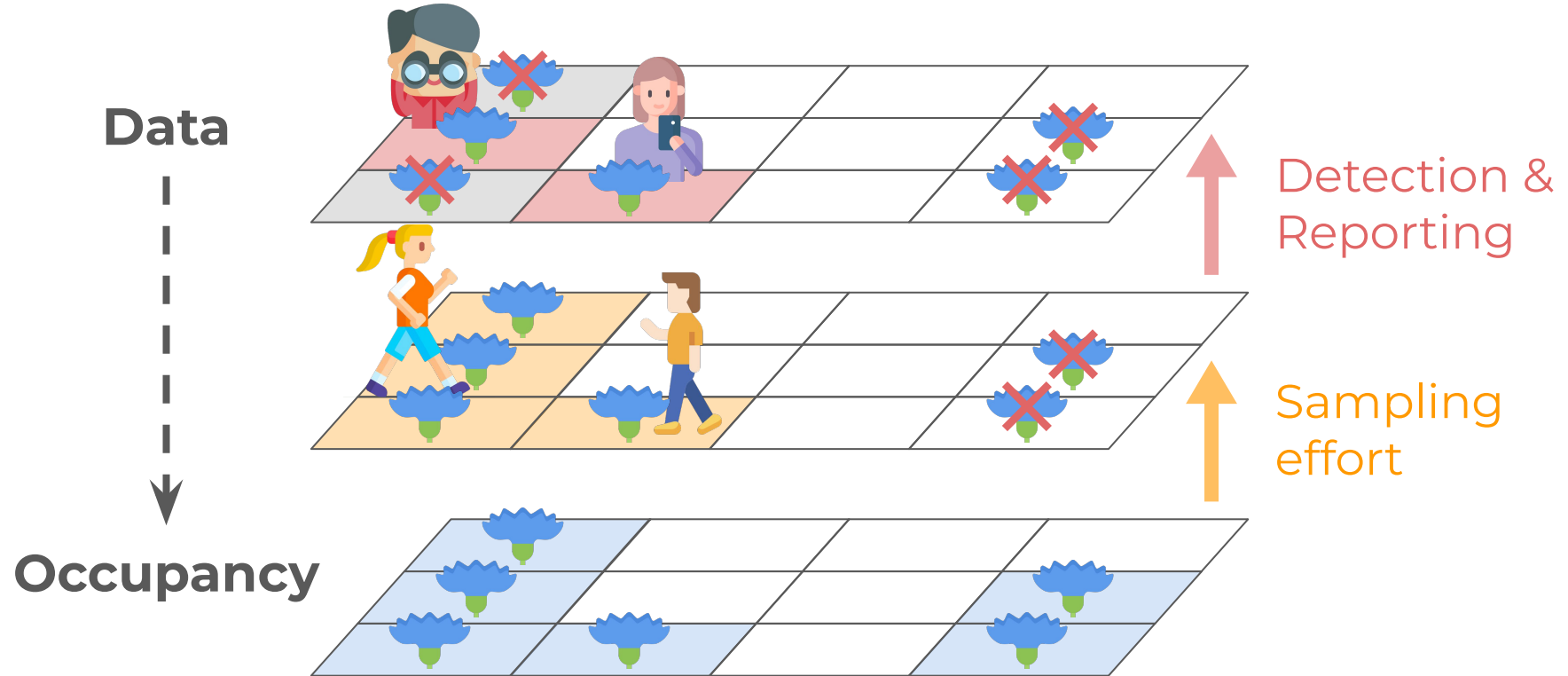
Occupancy



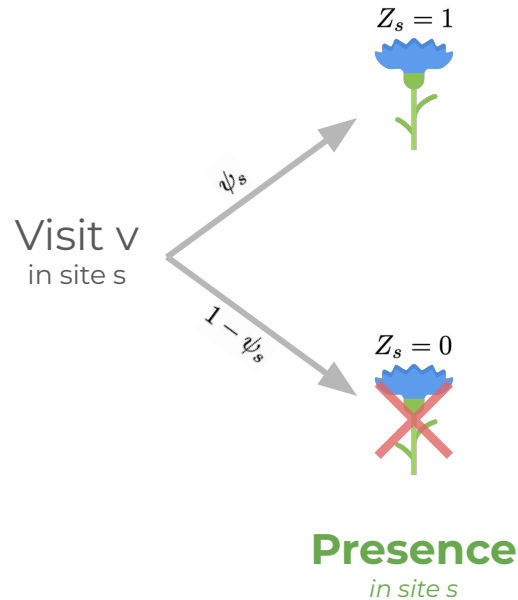
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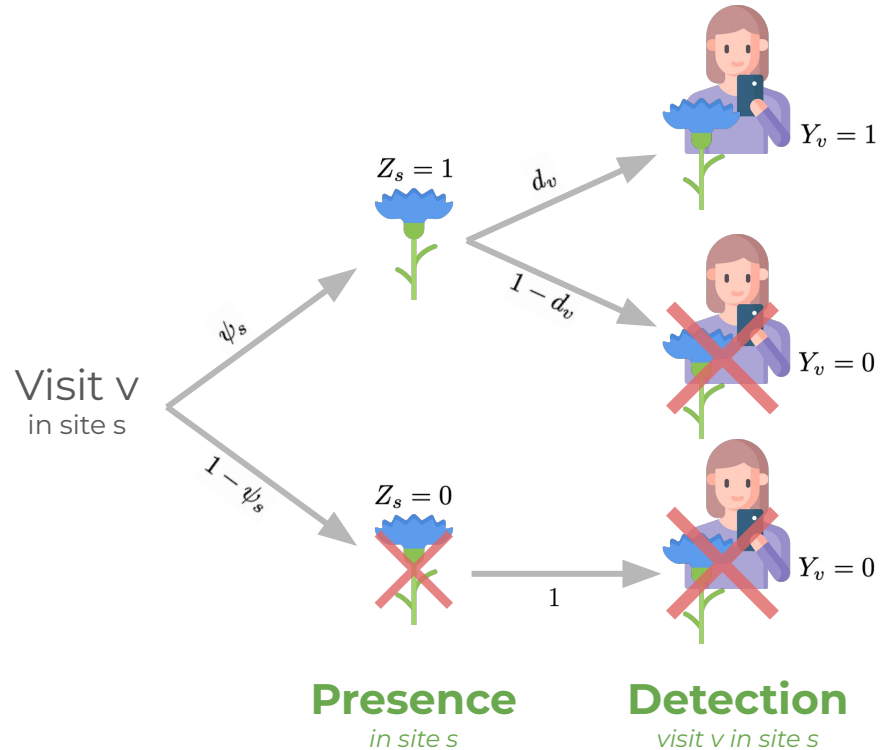
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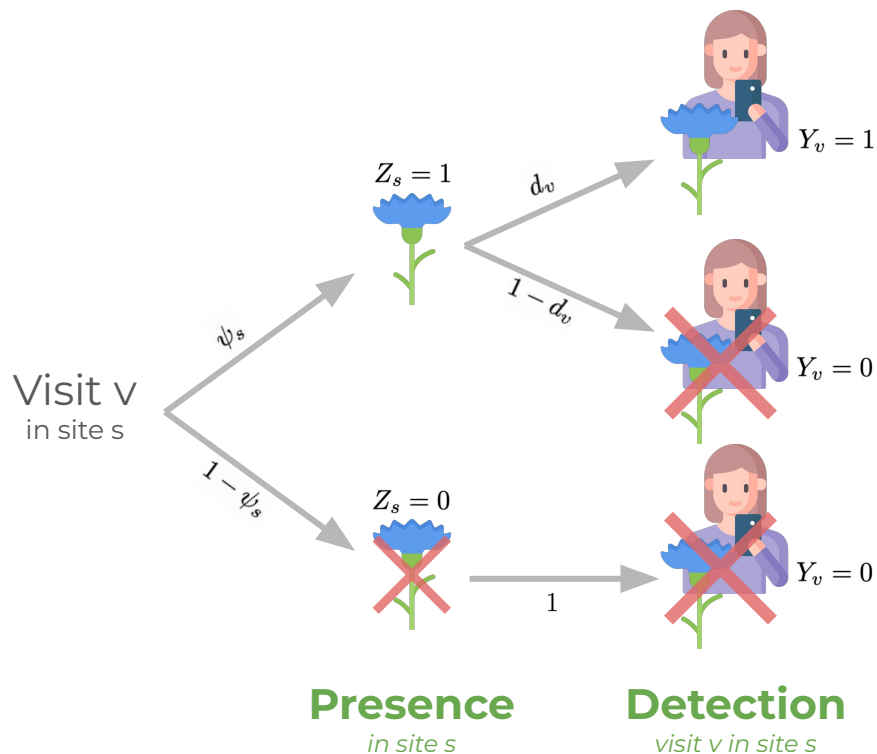
Single-season single-species



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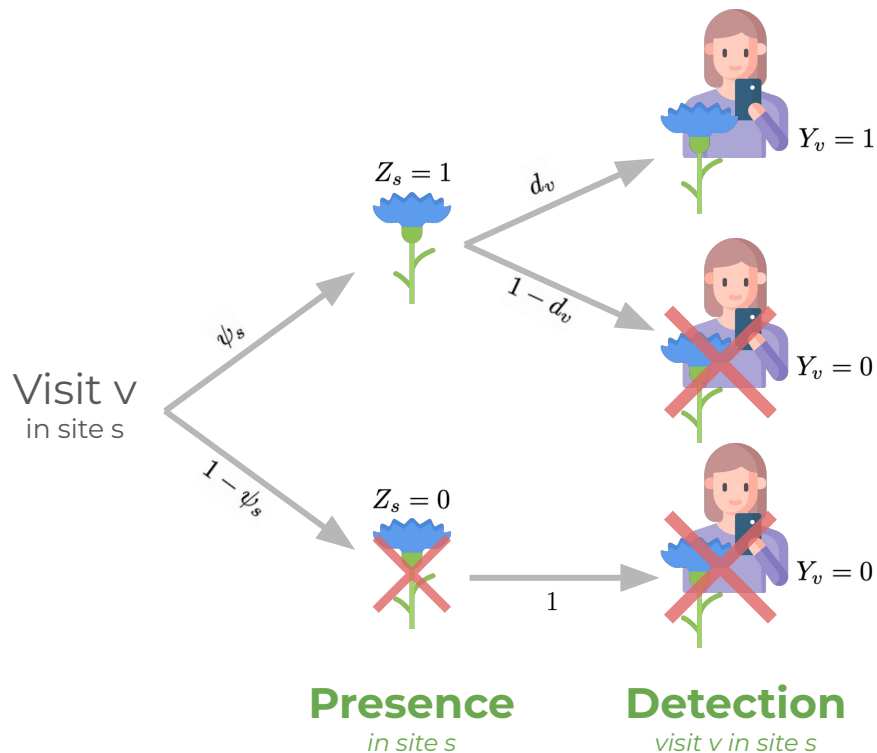


(Presence) $Z_s \stackrel{\text{ind.}}{\sim} \mathcal{B}(\psi_s)$

(Detection) $Y_v | Z_{s(v)} \stackrel{\text{ind.}}{\sim} \mathcal{B}(d_v \cdot Z_{s(v)})$

- **Explicit likelihood** to maximize
(only depending on ψ_s , d_v and y_v)

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- **Explicit likelihood** to maximize (only depending on ψ_s , d_v and y_v)
- Models for ψ_s and d_v could be:
 - Sigmoid of linear predictors
 - Sigmoid of **neural network**
 - ...
- **Multiple independent species:** product of the likelihood

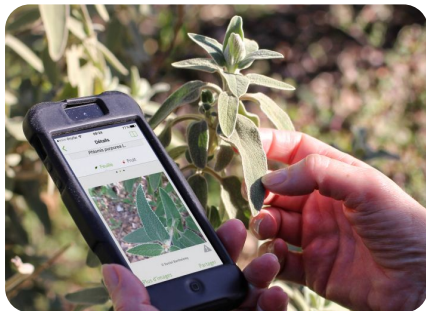
Citizen science programs - Opportunistic data

e.g.  Pl@ntNet

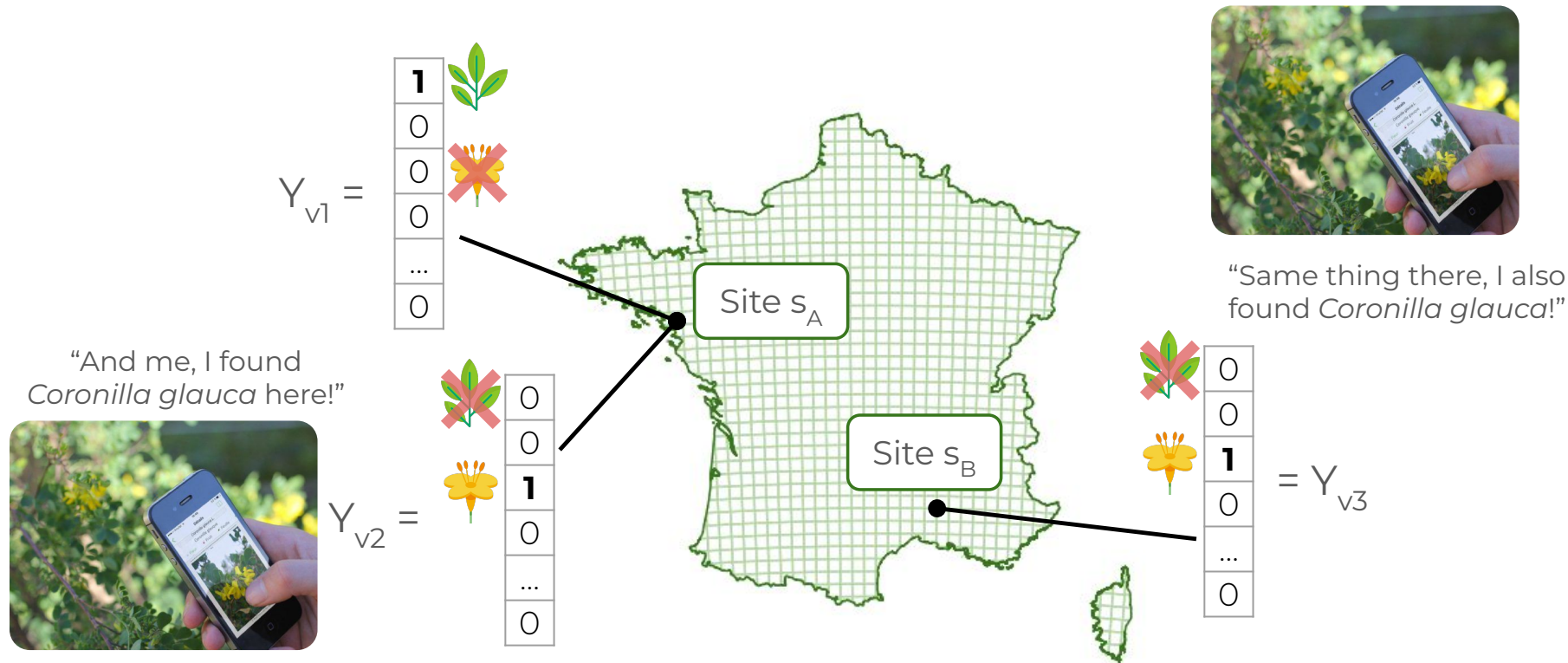
For K species

$$Y_{v1} = \begin{Bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{Bmatrix}$$


"Hey, I found *Phlomis purpurea* here!"



Citizen science programs - Opportunistic data



Challenges with these data

Massive datasets (e.g. a subset of **Pl@ntNet**):

- **2,605,648 observations**
- **12,289 species**

Unbalanced over space:

- Paris: 971.7 observations per km²
- Creuse: 1.1 observations per km²

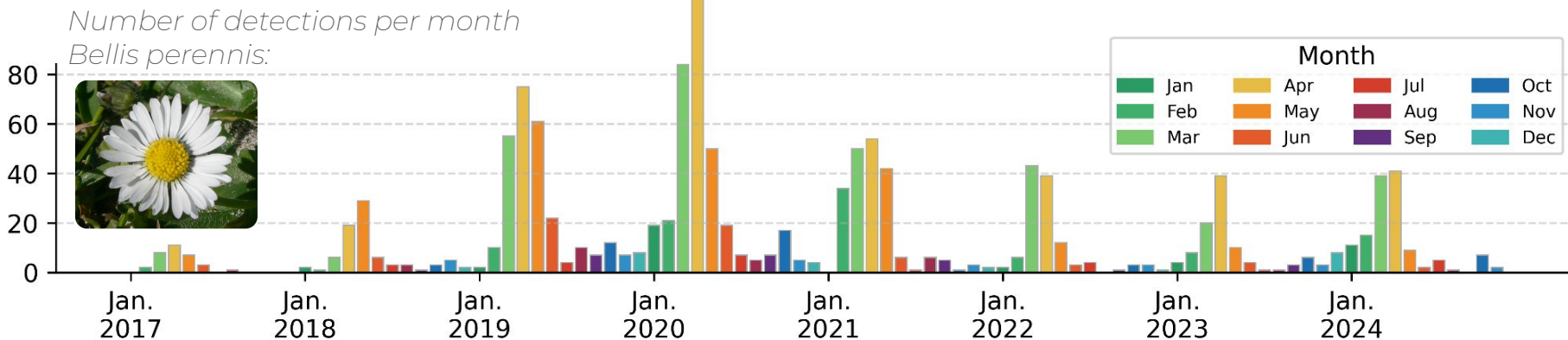
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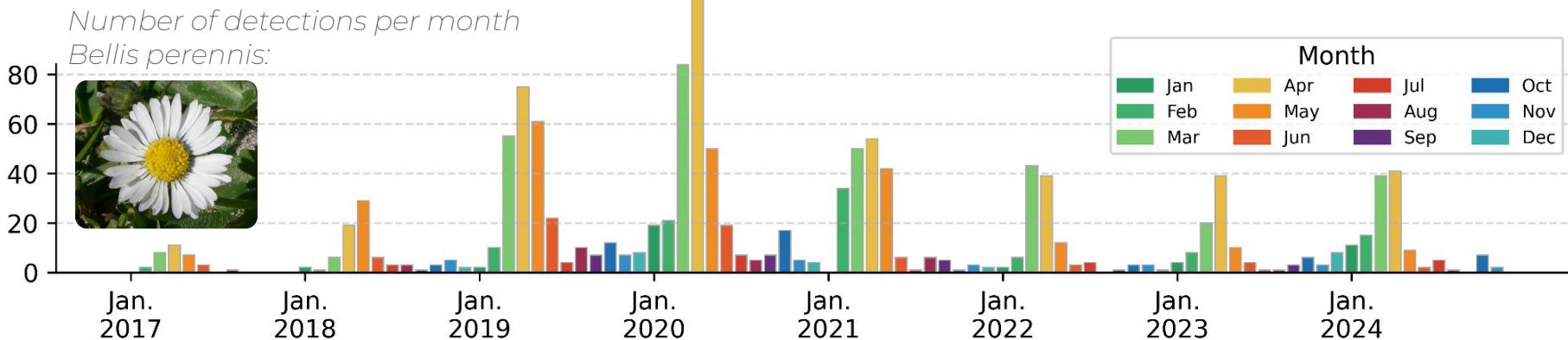
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→ Require **efficient** implementation
(data structure, optimizer...)

→ **Heterogeneous and complex** patterns
(non-linear effects, detection process...)



OccuPYcy

- Implemented in  python™, relying on  PyTorch, and inspired by the syntax of `unmarked` (R package)

OccuPYcy

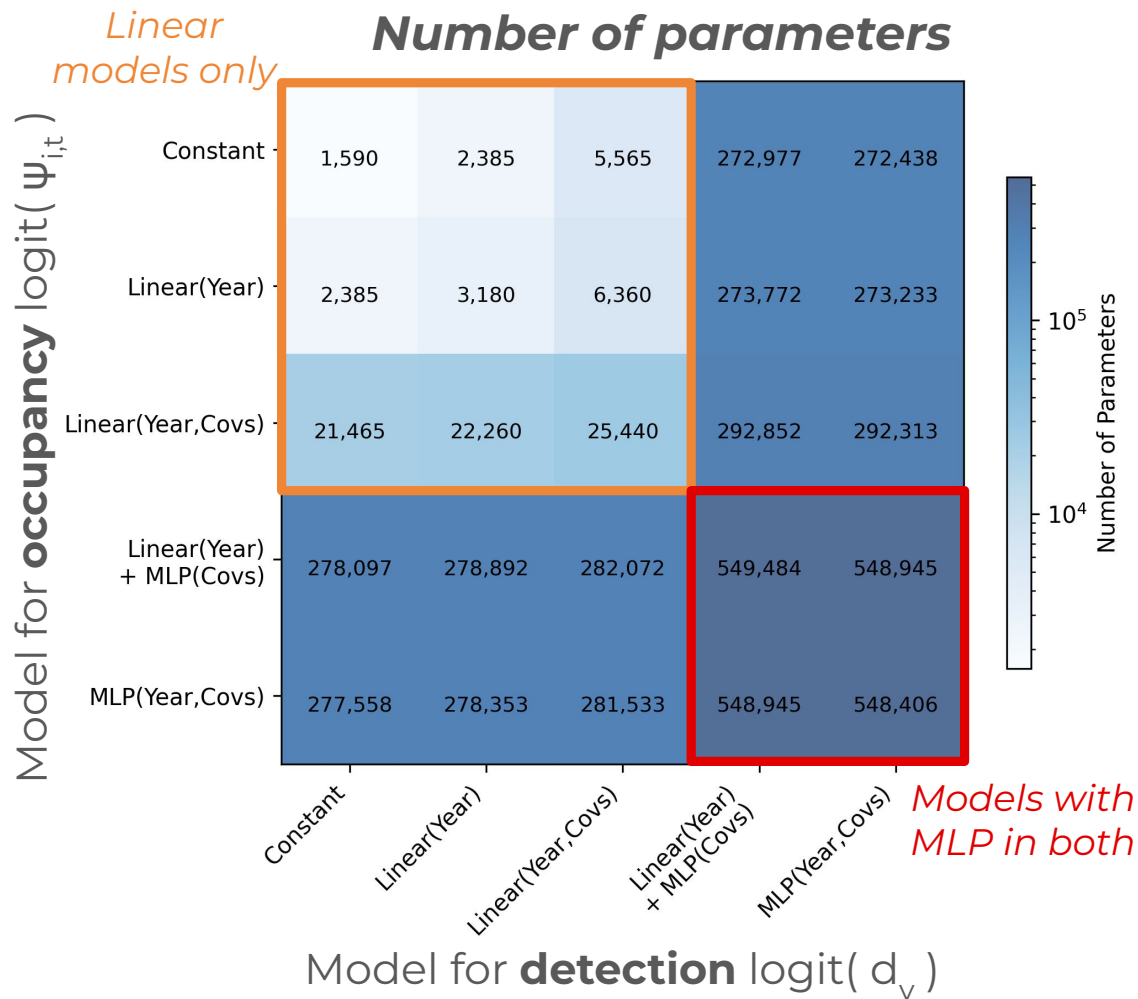
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 - LBFGS, SGD, Adam, ...
 - L2 / L1 regularizations on parameters / probabilities
 - Cross-validation routines for hyperparameter tuning

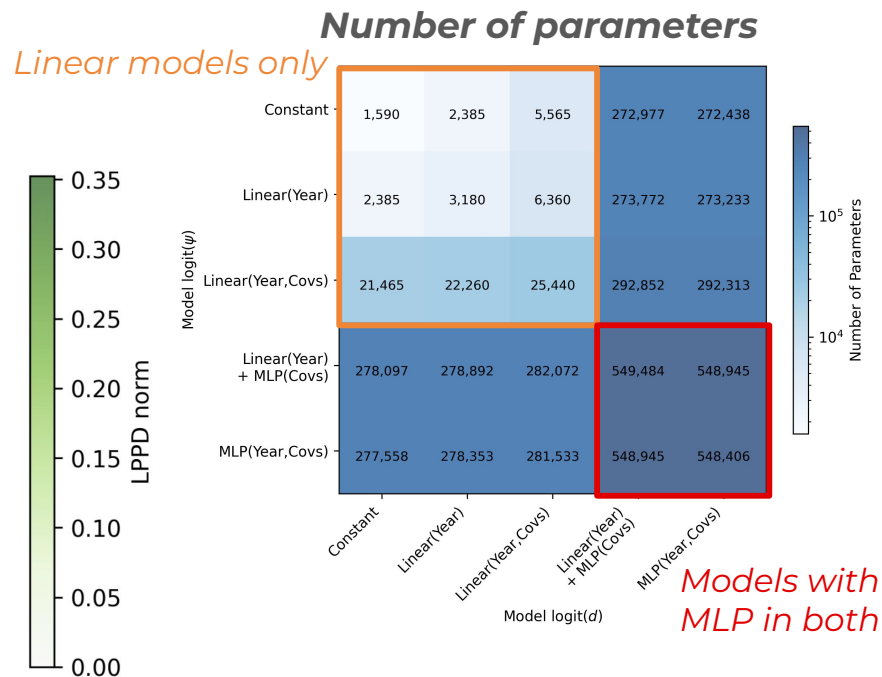
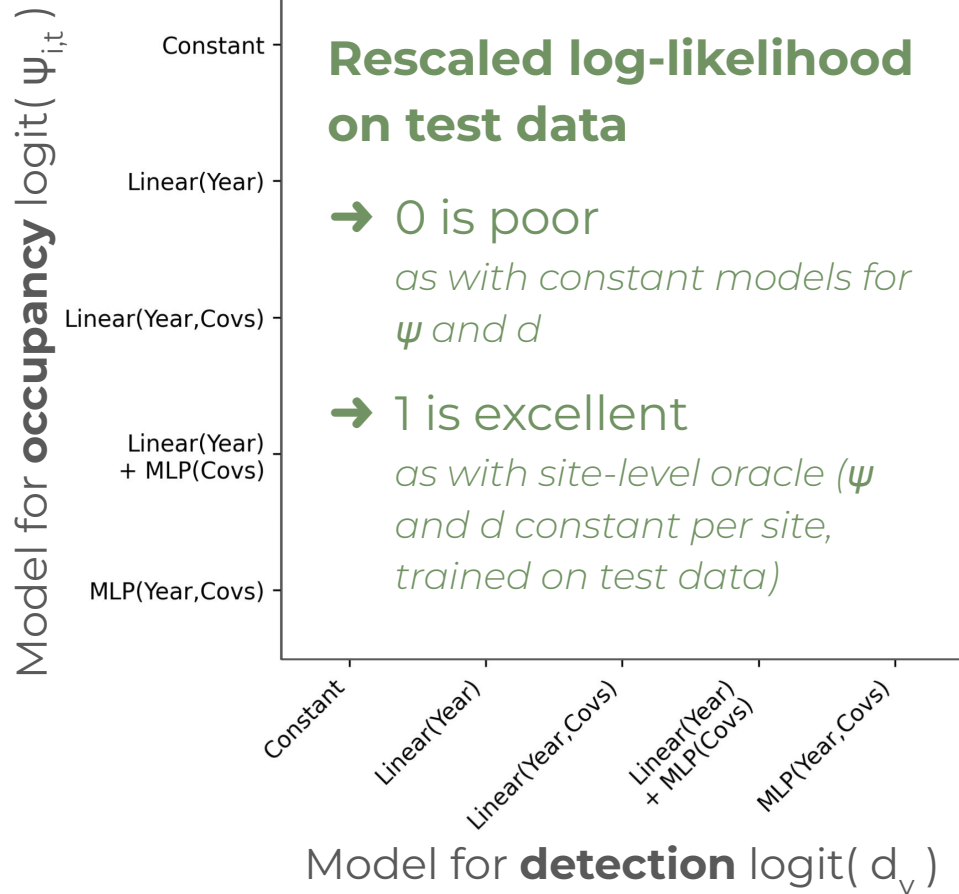
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- **Flexible models** for occupancy and detection
 - Predefined models – e.g. sigmoid of linear, sigmoid of neural networks...
 - Customisable: user-defined models from equation, formula, architectures...

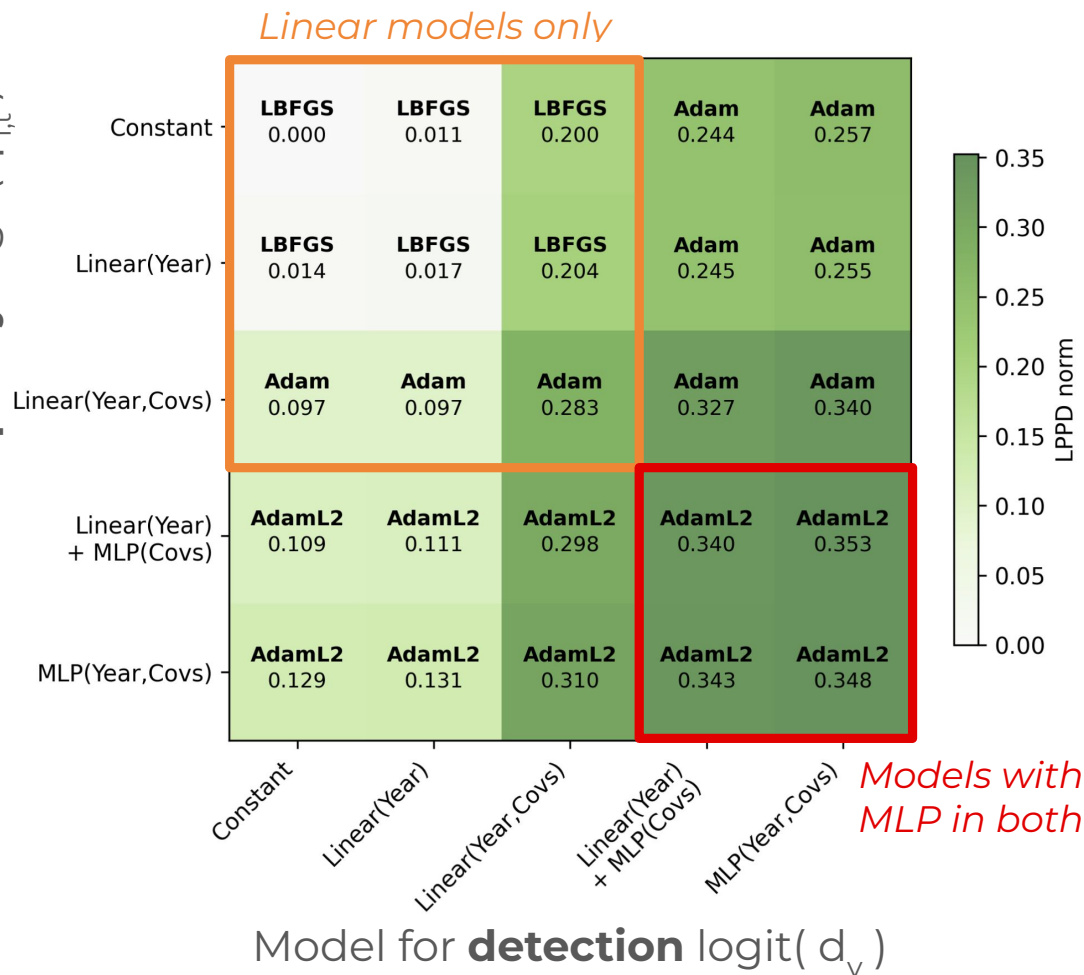
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 - Predefined models – e.g. sigmoid of linear, sigmoid of neural networks...
 - Customisable: user-defined models from equation, formula, architectures...
- Suitable **for large, heterogeneous datasets**
 - Sparse data structures, dataloaders, and stochastic optimization

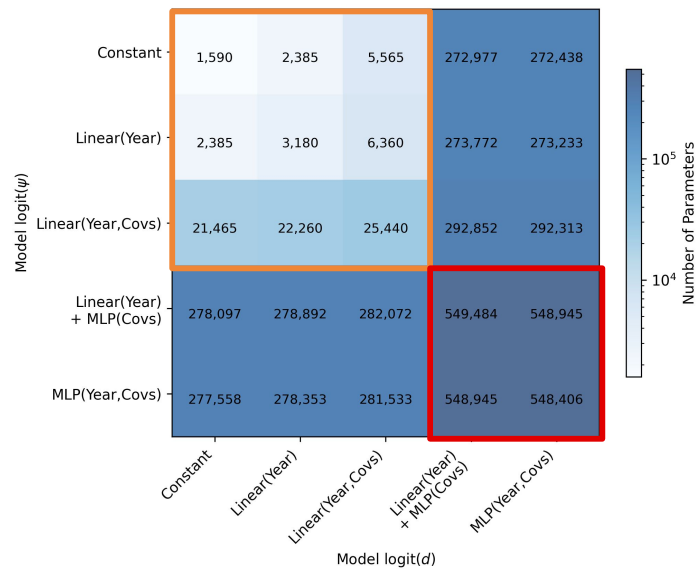




Model for **occupancy** $\logit(\psi_{i,t})$



Number of parameters

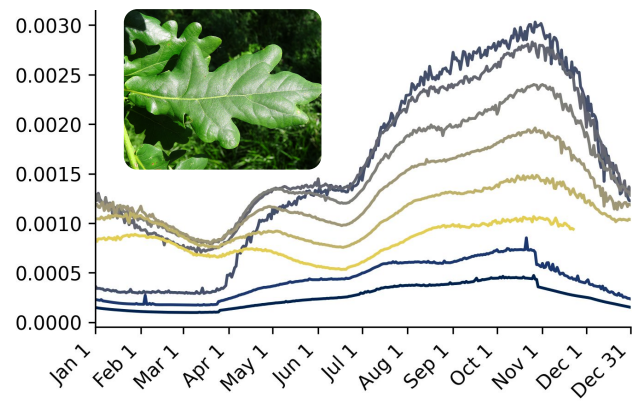


MLP models
“extrapolate” better
on test data

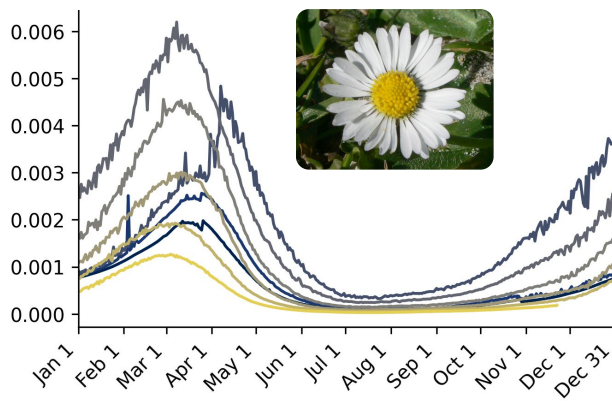
Fitted detection probability



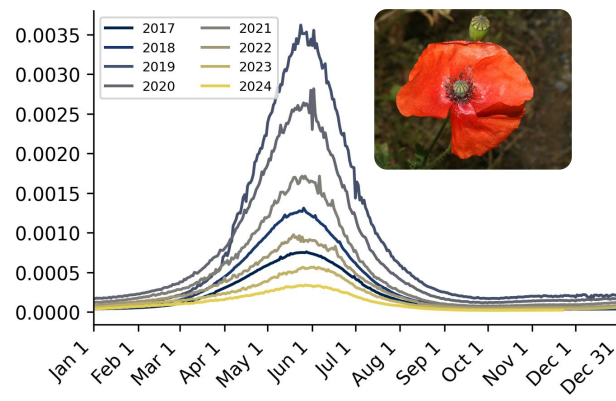
Quercus robur



Bellis perennis



Papaver rhoeas



Occupancy: $\text{logit}(\psi_{i,t}) = \text{coef} \times \text{year} + \text{MLP}(\text{bioclim}, \text{land use})$

Detection: $\text{logit}(d_v) = \text{MLP}(\text{year}, \sin\text{-cos Day of year}, \sin\text{-cos Time in day})$

Current and future directions

- Plant **trend estimation** using OccuPYcy on opportunistic data
Manuscript in preparation
- OccuPYcy package will be **released** soon (**end of June** 🙌)
Contact us if you're interested
Github URL when released: <https://github.com/RaphBnrd/OccuPYcy>
- **Ecological** interpretation of the results
- **Uncertainty** quantification on deep occupancy models

thanks for your attention

Supervisors:



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Bibliography

- [1] MacKenzie, D. I., Nichols, J. D., Lachman, G. B., Droege, S., Andrew Royle, J., & Langtimm, C. A. (2002). Estimating Site Occupancy Rates When Detection Probabilities Are Less Than One. *Ecology*, 83(8), 2248-2255. [https://doi.org/10.1890/0012-9658\(2002\)083%255B2248:ESORWD%255D2.0.CO;2](https://doi.org/10.1890/0012-9658(2002)083%255B2248:ESORWD%255D2.0.CO;2)
- [2] Guillera-Aroita, G. (2017). Modelling of species distributions, range dynamics and communities under imperfect detection : Advances, challenges and opportunities. *Ecography*, 40(2), 281-295. <https://doi.org/10.1111/ecog.02445>
- [3] Joseph, M. B. (2020). Neural hierarchical models of ecological populations. *Ecology Letters*, 23(4), 734-747. <https://doi.org/10.1111/ele.13462>